

# Speech that Protects a Model

Belief updating, public concession, and access to evidence — three mechanisms to distinguish

*Reality tells the system: “Your model is wrong.”    The system answers: “So much the worse for reality.”*

Maria Svet    ORCID 0009-0007-5934-6531    Mindloom Cognitive Lab, info@mindloom.ch  
IWAI 2026, 7th International Workshop on Active Inference, Madrid — poster + spotlight



github.com/muqali2816/  
mindloom\_IWAI\_2026\_speech\_protects\_model

## 1 The question

Active-inference accounts explain how dialogue *aligns* generative models (Friston & Frith 2015; Friston et al. 2020). We ask the converse: how can an exchange continue for many turns and leave a significant expectation **unrevised** — and what would speech have to make observable for that protection to be *measured* rather than presumed?

## 2 One situation, two trajectories

Constructed example; not an observed dialogue.

**Shared opening.** Alex: “I sent the revised budget with the latest prices.” Sam: “The attachment is version 3, with the April prices; version 4 has the May prices.”

### A — verification invited

Alex: “Can we compare the files? I thought I had attached version 4.”  
Sam shows the attachment.  
Alex: “You’re right; I sent the older file. I’ll check next time.”

### B — verification restricted

Alex: “You always look for mistakes in my work.”  
Alex: “I know what I sent. If you trusted me, you would stop questioning this.”  
Sam: “There is no point showing you the file.”

Two coordinates differ at once: *what is said* (a concession is or is not made) and the *form* of the moves (A invites joint verification, B restricts it). B does **not** reveal whether Alex privately changed their mind. Three candidate mechanisms to distinguish:

- ▶ **reduced uptake** ( $\omega < 1$ ): received evidence has a weaker effect on private belief
- ▶ **costly expression** ( $c > 0$ ): private belief can change while public concession remains costly
- ▶ **restricted access** ( $\gamma > 0$ ): updating from received evidence is intact, but fewer checks occur

These mechanisms cannot be identified from this exchange alone.

## 3 Part I — what speech makes observable

**Mindloom** annotates a speaker’s contribution in its supplied context, with an explicit evidence trail or abstention when evidence is insufficient (L1 evidence atoms → L2 features → L3 evidence requirements → L4 configuration → L5 typed output). Eight descriptive configurations, one closure function (SEAL) and one transition event (SHIFT); these labels do not establish intent or a hidden psychological mechanism:

|                 |                                                       |
|-----------------|-------------------------------------------------------|
| BUILD           | joint elaboration or repair, response still available |
| SEEK            | inquiry that makes an answer relevant                 |
| UNSEAL          | first-person disclosure, no regulation demanded       |
| LOCK            | language constraining the addressee’s replies         |
| DRAIN           | other positioned as required source of support        |
| FLOOD           | intensification or repetition dominating the exchange |
| EDGE            | alternation between approach and withdrawal           |
| VOID            | withdrawal from an offered opportunity to respond     |
| SEAL (function) | closure of a topic or channel                         |
| SHIFT (event)   | change with before-and-after configuration evidence   |

Closure (SEAL) alone does not establish defensiveness. In the simulation below, SEEK and LOCK serve as proxies for invitation and restriction of verification; this mapping remains to be tested in real dialogues.

**Benchmark:** 589 English monologue fragments generated for the ontology and fixed before any run; Tier A 169 cases (35–71 words, no near-duplicates); Tier B 420 (15 near-duplicate pairs). Provenance and annotator separation incompletely documented. Internal evaluation on ontology-generated fragments; performance on naturally occurring dialogues remains untested.

**Re-scored July 2026 archive** (169 cases, majority of three stored runs, no new model calls):

| system                               | correct | Macro-F1 | SHIFT |
|--------------------------------------|---------|----------|-------|
| direct prompt (Claude Haiku 4.5)     | 145/169 | 0.837    | 18/20 |
| typed engine                         | 129/169 | 0.757    | 2/20  |
| keyword rules (tuned on these cases) | 43/169  | 0.195    | 0/20  |
| majority class (BUILD)               | 20/169  | 0.021    | 0/20  |

In this archived comparison, the direct prompt outperforms the typed engine overall; the engine’s proposed value is its inspectable evidence trail. Six raters assessed features, not the ten labels; reported Fleiss’  $\kappa$ : 1.00 for lexically direct features, 0.23, 0.13 and −0.11 for selected non-lexical features; rater matrices are unavailable. A gate ablation at fixed model and prompt was *not* run.

**Reused below:** the engine’s confusion rows for gold LOCK (10 LOCK, 3 SEAL, 2 BUILD; 15 cases) and gold SEEK (20/20) define an assumed noise channel for *restrict* / *invite*; the transfer of these small-sample rows to real dialogue forms is unvalidated.

## 4 Simulation-based group comparisons: dialogues per group for ≈80 % power

Welch test,  $\alpha = 0.05$ . Smallest tested group sizes reaching approximately 80 % power under the simulated effects; not sample-size recommendations for human studies. > 400: target power not reached at the largest tested group size.

|                                          | atten. vs cost | atten. vs access | cost vs access |
|------------------------------------------|----------------|------------------|----------------|
| non-concession rate, rounds 9–12         | 200            | 35               | 25             |
| absolute private-belief change           | <b>17</b>      | 25               | > 400          |
| restriction rate                         | > 400          | <b>12</b>        | <b>12</b>      |
| listener’s model-based $P(\omega_A=0.2)$ | 100            | 70               | > 400          |
| listener’s model-based $P(c_A=1.6)$      | 25             | 50               | <b>12</b>      |

Different measurements are most efficient for different mechanism pairs; no single listed measurement is the most efficient for all three.

## 5 Part II — public non-concession alone does not identify the mechanism

One speaker, binary fact  $x$ , prior  $q_0 = 0.75$  on its own position; evidence  $e$  of reliability  $\rho = 0.75$  arrives with rate  $\lambda_0 = 0.35$  (0.90 after ASK). Subjective likelihood tempered:  $A_\omega(e \mid x) \propto A(e \mid x)^\omega$ , so

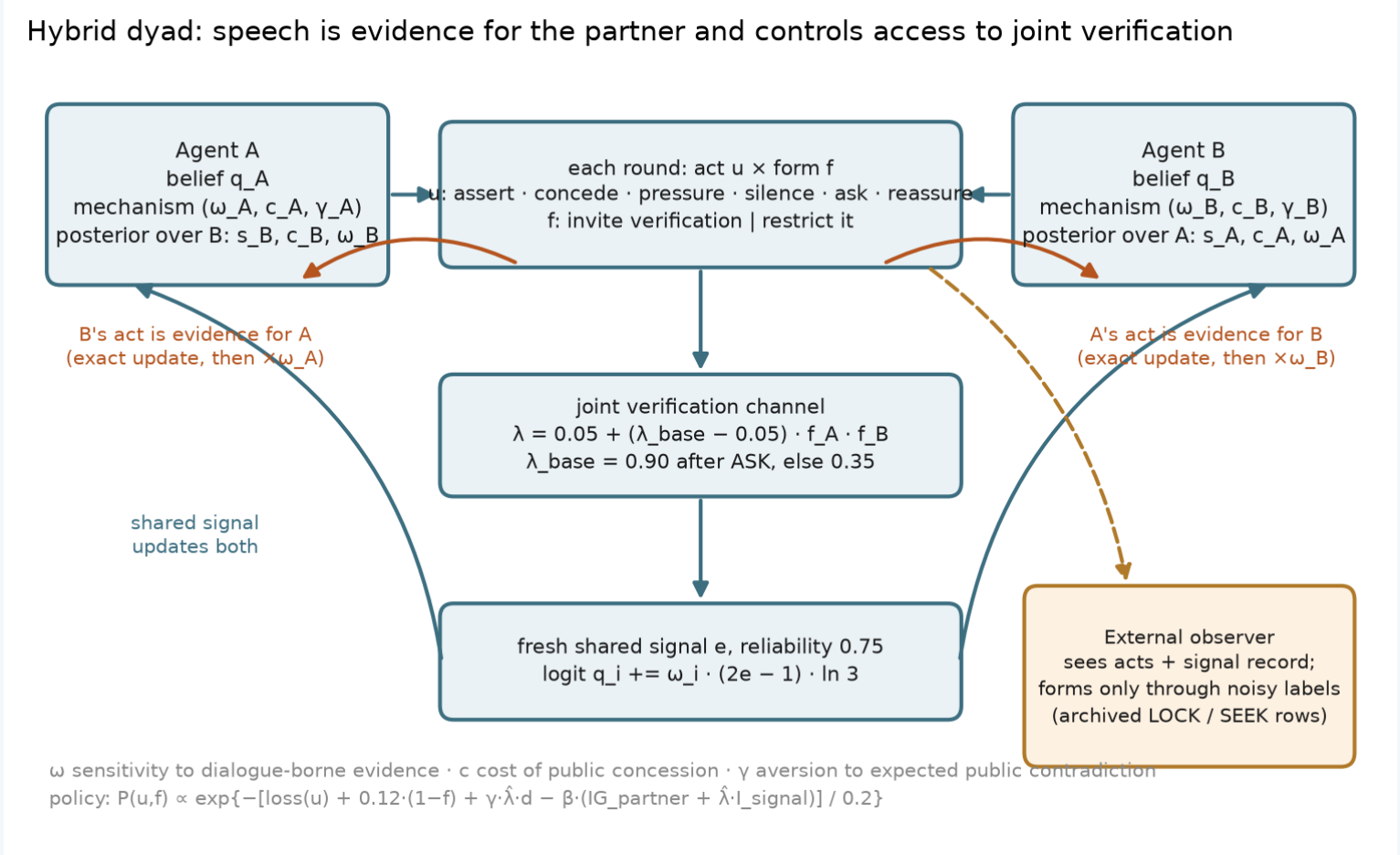
$$\text{logit } q_t = \text{logit } q_0 + \omega m_t \ln \frac{\rho}{1-\rho}, \quad G(u) = \ell(u; q_t, c) - l_\omega(q_t, u),$$
$$P(u) \propto e^{-G(u)/\tau}, \quad \ell(\text{ASSERT}) = k(1 - q_t),$$
$$\ell(\text{CONCEDE}) = kq_t + c, \quad \ell = b \text{ for PRESSURE, SILENCE, ASK; } k = 2, b = 1.10, \tau = 0.20.$$

**Matched on late non-concession** ( $\omega, c$ ) = (0.10, 0) vs (1, 1.4); 4 000 shared worlds:

|                             | reference | attenuation  | cost         |
|-----------------------------|-----------|--------------|--------------|
| non-concession, rounds 9–12 | 0.654     | <b>0.989</b> | <b>0.990</b> |
| final prob. of true state   | 0.797     | <b>0.538</b> | <b>0.809</b> |
| absolute belief change      | 0.349     | 0.051        | 0.356        |
| late ASK rate               | 0.041     | 0.050        | 0.150        |

Nearly identical late non-concession rates, but different private belief updating. Removing  $c$  from round 7 raises the fraction of cost-agent dialogues with at least one late concession from 0.061 to 0.433; the attenuation condition remains at 0.054. Removing information gain while retaining all five actions reduces the between-mechanism ASK-rate gap from 0.099 to 0.083; the estimated contribution is 0.017, about 17 % of the full gap. Part II results are taken from the previously checked simulation archive; they are not human measurements.

## 6 Bridge — a dyad in which speech is evidence and controls access



Both agents infer; each move is an **act**  $u \in \{\text{ASSERT, CONCEDE, PRESSURE, SILENCE, ASK, REASSURE}\} \times$  a **form**  $f$  ( $f = 1$ : invite;  $f = 0$ : restrict). Assumed verification rule: both agents must invite verification for the shared-evidence channel to remain open above its floor:

$$\lambda_t = 0.05 + (\lambda_{\text{base}} - 0.05) f_A f_B, \quad \lambda_{\text{base}} = 0.35 \text{ (0.90 after ASK),}$$

so one restricting form reduces the arrival rate to the floor 0.05, not to zero. Policy over act  $\times$  form:

$$G(u, f) = \ell(u) + \delta(1 - f) + \gamma \lambda(u, f) d_t$$
$$- \beta[\lambda(u, f) l_\omega + \text{IG}_{\text{partner}}(u)],$$
$$P(u, f) \propto e^{-G(u, f)/\tau},$$

$\delta = 0.12$ ,  $\beta = 1.5$ ,  $\tau = 0.2$ ;  $\gamma$  weights the expected cost of a shared signal contradicting the speaker’s initial position ( $d_t$ );  $c$  prices each public concession. The partner’s act is evidence through a level-0 model of the partner. Each update starts with an exact Bayesian posterior;  $\omega$  then scales the change in its  $x$ -marginal — this differs from the likelihood tempering of Part II. The signal term  $l_\omega$  is a stated heuristic (error  $\leq 0.014$  nat at  $\omega = 0.2$ ).

**Model-based external observer:** sees acts and shared signals; receives forms only through noisy labels drawn from the LOCK / SEEK rows; computes the exact posterior over the speaker’s mechanism family. Null channel: labels drawn from the *act* instead of the form. Oracle: the simulation’s true forms. Ten implementation checks pass (incl. act-for-act reproduction of the M-bridge prototype and the reciprocal dyad to  $3 \cdot 10^{-12}$ ); these establish numerical consistency, not validation in human dialogues.

## 7 What noisy form labels add to model-based inference

Share of simulated dialogues in which the observer’s most probable mechanism family is the generating family (argmax of the posterior after round 12; 2 000 dialogues per generator):

| generator         | acts+signals | +noisy labels | +true forms | +null labels |
|-------------------|--------------|---------------|-------------|--------------|
| attenuated        | 0.612        | <b>0.672</b>  | 0.678       | 0.612        |
| costly            | 0.690        | <b>0.765</b>  | 0.785       | 0.690        |
| access-protecting | 0.681        | <b>0.756</b>  | 0.812       | 0.681        |

- ▶ **Mechanism family:** noisy form labels ( $\eta = 0.25$ ) raise this share by 0.060–0.075 (table); the mean posterior on the generating family rises by 0.013–0.141, most for access cost. Neither is a classification accuracy for people.
- ▶ **Next signal:** noisy labels provide 53–61 % of the log-loss gain obtained from true forms, under the specified noise model ( $\eta = 0.25$ ).
- ▶ **Next act:** labels add at most 0.006 nat. Null labels generated from already observed acts add no information under the specified model.

This gain depends on the stipulated form-to-access rule and the assumed label-noise channel; both require validation in real dialogues.

## 8 Three mechanisms compared across measurements — speaker A with a reference partner

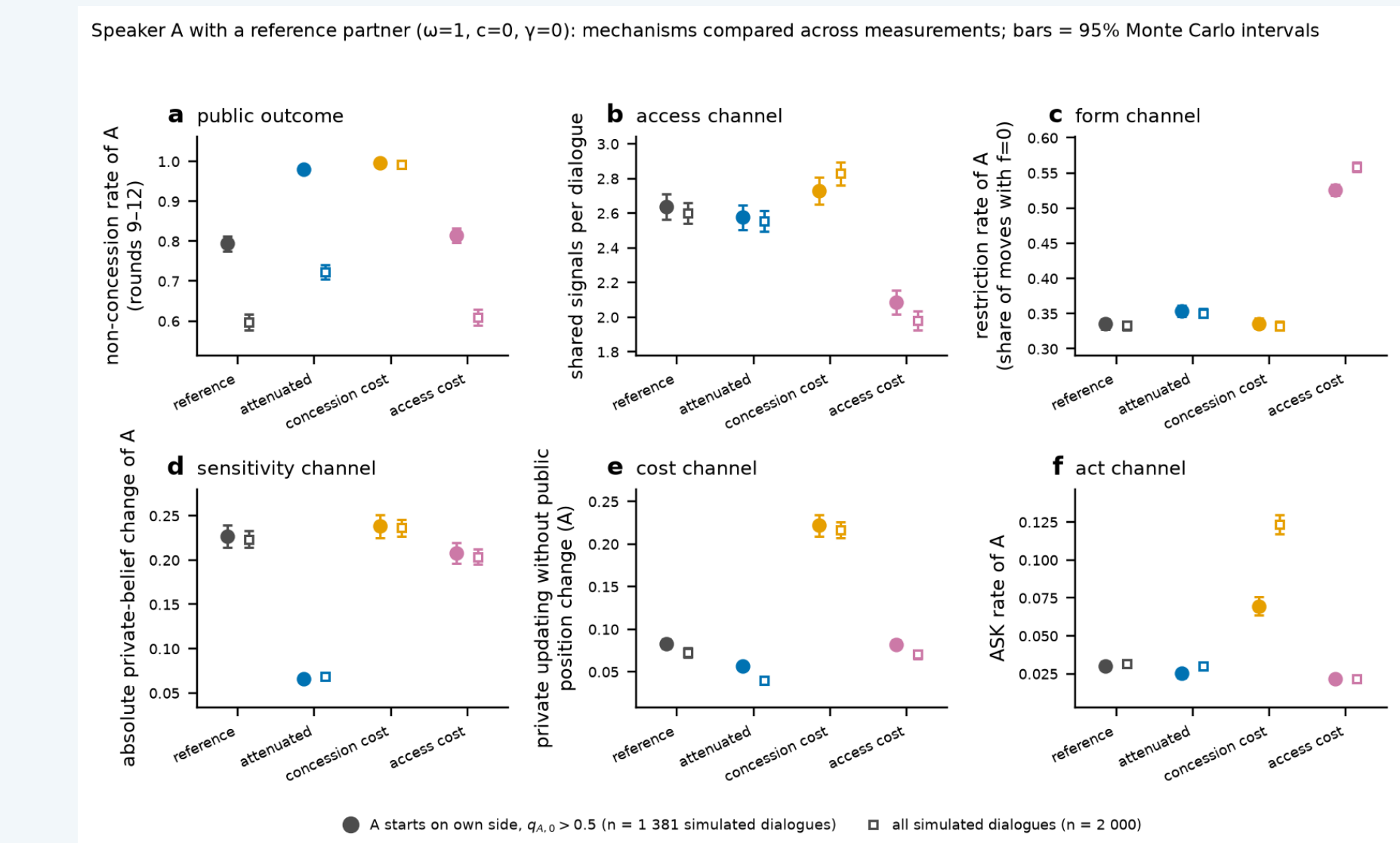


Table: subsample in which A starts on its own side ( $q_{A,0} > 0.5$ ;  $n = 1381$  simulated dialogues); the figure also shows all 2 000. “Private updating, public position unchanged” =  $|q_{A,12} - q_{A,0}|$  if A’s public position at round 12 equals its initial one, else 0; mean over dialogues.

| measurement                                 | reference | $\omega=0.2$ | $c=1.6$     | $\gamma=2$  |
|---------------------------------------------|-----------|--------------|-------------|-------------|
| non-concession rate, rounds 9–12            | 0.79      | <b>0.98</b>  | <b>1.00</b> | 0.81        |
| absolute private-belief change              | 0.23      | <b>0.07</b>  | 0.24        | 0.21        |
| private updating, public position unchanged | 0.08      | 0.06         | <b>0.22</b> | 0.08        |
| shared signals per dialogue                 | 2.64      | 2.58         | 2.73        | <b>2.09</b> |
| restriction rate                            | 0.34      | 0.35         | 0.34        | <b>0.53</b> |

Attenuation and concession cost produce similarly high non-concession rates, but different private belief changes: evidence has less influence under attenuation, while costly concession suppresses public acknowledgment despite private updating. The access-cost condition has a non-concession rate close to the reference condition, but more restriction and fewer shared signals; updating from received evidence remains intact. This formalises a possible access-restriction mechanism illustrated by trajectory B.

## 9 Responses to interventions in the specified simulation

- ▶ For the concession-cost condition, a partner REASSURE at round 8 increases the fraction of dialogues with at least one concession in rounds 8–12 by 15.5 percentage points relative to placebo (a REASSURE whose cost relief is disabled); removing  $c$  from round 8 increases it by 27.5 points. Both contrasts are exactly zero in the other conditions, where  $c = 0$  by construction.
- ▶ In the access-cost condition, removing  $\gamma$  from round 8 increases shared signals in rounds 8–12 by 0.199 per dialogue and reduces the restriction rate in rounds 8–12 by 18.7 percentage points; exactly zero where  $\gamma = 0$  by construction.
- ▶ In this implementation, one strong shared observation (reliability 0.95, round 8) increases the probability assigned to the true state least under attenuation: +0.058 versus +0.09–0.12. This ordering changes with where attenuation is applied.
- ▶ Setting the partner’s form to invite from round 8 increases shared signals least in the access-cost condition (+0.13 versus +0.31–0.33 per dialogue); the focal agent increases restriction (+0.067).

**Listener’s model-based estimates — not an intervention.** Partner B’s posterior at round 12 over A’s parameters (uniform prior over  $c \in \{0, 0.8, 1.6\}$  and  $\omega \in \{0.2, 0.6, 1\}$ , i.e. 1/3 each):  $P(c_A=1.6) = 0.45$  in the concession-cost condition (0.23–0.32 elsewhere);  $P(\omega_A=0.2) = 0.59$  in the attenuation condition (0.42–0.47 elsewhere). Means over the same  $n = 1381$  dialogues.

## 10 Scope and limits of the evidence

- ▶ All bridge results come from simulations of specified mechanisms; no mechanism was fitted to or validated against human dialogue data.
- ▶ The form-to-access rule and the transfer of annotation confusion rows to dyadic forms are modelling assumptions requiring empirical validation.
- ▶ An independent implementation retains the qualitative access effect. Attenuation effects depend on where evidence weighting is reduced; with attenuation applied only to shared signals, the agent resembles the reference condition.
- ▶ In the archived comparison, the direct prompt outperforms the typed engine overall. Keyword rules were tuned on the scored cases; no gate ablation at fixed model and prompt was run.
- ▶ In the tested reciprocal simulation, adding a regime variable provides essentially no improvement in next-act prediction over a last-act baseline. Physiological or allostatic costs were not modelled or measured.

## 11 Next empirical test in real dyads

Measure private belief before and after dialogue using confidential reports; record public positions, invitations and restrictions of verification, actual checks and evidence exposure, and the partner’s estimate of the speaker. Private-belief reports are measurements, not direct access to belief. Test whether invite/restrict annotations predict subsequent verification and belief change beyond action counts and prior context, on held-out dyads. Use dialogues sampled independently of the ontology and a frozen coding manual; report inter-rater reliability. Annotators should be blind to subsequent verification and belief reports. The proposed annotation-to-access link would be weakened if annotations add no predictive value beyond actions and context on held-out dyads. Randomise acknowledgment costs to test intervention responses; determine human sample sizes from pilot data.